Gradient Descent and Optimization

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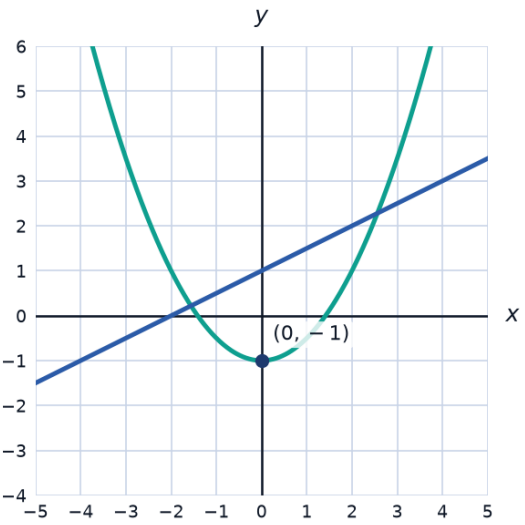
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Consider the following quadratic loss function for a regression problem: $L(y, \hat{y}) = (y - \hat{y})^2$. If the predicted value $\hat{y} = 5$ and the true value $y = 3$, what is the value of the loss function?

2 [Foundational] • Target: Loss Function Derivatives and Stationary Points

Consider a discrete random variable X with the following probability mass function (PMF): $P(X = x) = \frac{x}{10}$ for $x \in \{1, 2, 3, 4\}$. What is the value of $P(X \leq 2)$?



3 [Foundational] • Target: Multivariate Gradients and Directional Derivatives

The sigmoid activation function is $\sigma(z) = \frac{1}{1 + e^{-z}}$. If $\sigma(z_0) = 1.20$, what is the value of the derivative $\sigma'(z_0)$?

4 [Foundational] • Target: Gradient Descent Step Updates

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.30$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

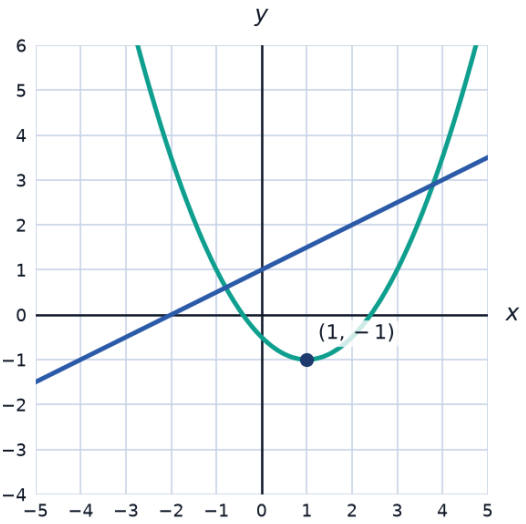
5 [Foundational] • Target: Momentum and Advanced Optimizers

For a bivariate feature dataset, $\text{Cov}(x, y) = 36$, $\text{Var}(x) = 12$, $\bar{x} = 5$, and $\bar{y} = 35$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

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6 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In training a language model token predictor, a binary token occurs $k = 15$ times across $n = 50$ independent trials. What is the Maximum Likelihood Estimate $\hat{p}_{\text{MLE}}$ of the token probability?



7 [Foundational] • Target: Loss Function Derivatives and Stationary Points

A spam classification model identifies $TP = 120$ genuine spam emails and $FP = 40$ legitimate emails falsely flagged as spam. What is the precision of this model?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Multivariate Gradients and Directional Derivatives

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.25$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

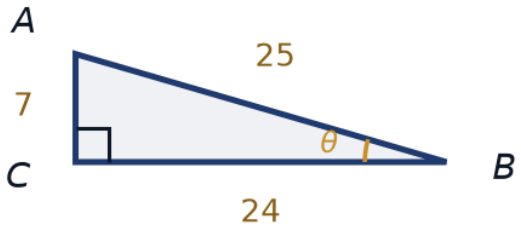
9 [Intermediate] • Target: Gradient Descent Step Updates

In anomaly detection, anomaly prior $P(A) = 0.02$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A| +)$ that an alarm is a true anomaly?

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10 **[Intermediate]** • **Target: Momentum and Advanced Optimizers**

Compute the Manhattan (L_1) norm and Euclidean (L_2) norm of the word vector $\mathbf{w} = [10, -15]^T$.



11 **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

Consider a simple quadratic loss function $L(w) = (w^2 - 4w + 5)$. We want to find the minimum of this loss function using gradient descent. The gradient of the loss function with respect to w is given by $\frac{dL}{dw} = 2w - 4$. If we start at an initial weight $w_0 = 5$ and use a learning rate $\alpha = 0.5$, what will be the value of the weight w_1 after the first update step, i.e., $w_1 = w_0 - \alpha \frac{dL}{dw}(w_0)$? Show your complete algebraic work.

12 **[Intermediate]** • **Target: Loss Function Derivatives and Stationary Points**

Consider a univariate regression model where the predicted output $\hat{y}$ is given by $\hat{y} = ax + b$. The loss function is the Mean Squared Error (MSE), defined as $L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$. $*For a single data point* (x_1, y_1) = (2, 5), *the loss is* L = (5 - (2a + b))^2. *The gradient of the loss function with respect to a is* \frac{\partial L}{\partial a} = -2(5 - (2a + b)) \cdot 2. *If during gradient descent, with a learning rate of* \eta = 0.1, *the parameters* a *and* b *are updated from* a_{old} = 1 *and* b_{old} = 1, *to* a_{new} = 0.8 *and* b_{new} = 0.8, *what is the value of* b *that would have resulted in* \frac{\partial L}{\partial b} = 0 *for this single data point, assuming* a=1 *and* b=1 *were the values * before * the gradient calculation that led to the update to* a_{new}=0.8, b_{new}=0.8$? Show your complete algebraic work.

13 **[Intermediate]** • **Target: Multivariate Gradients and Directional Derivatives**

Consider the scalar loss function $L(w) = 6w^2 - 8w + 5$. Find the analytical gradient $abla L(w)$ evaluated at $w = 10$.

14 **[Intermediate]** • **Target: Gradient Descent Step Updates**

Consider the scalar loss function $L(w) = 7w^2 - 8w + 5$. Find the analytical gradient $abla L(w)$ evaluated at $w = 12$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

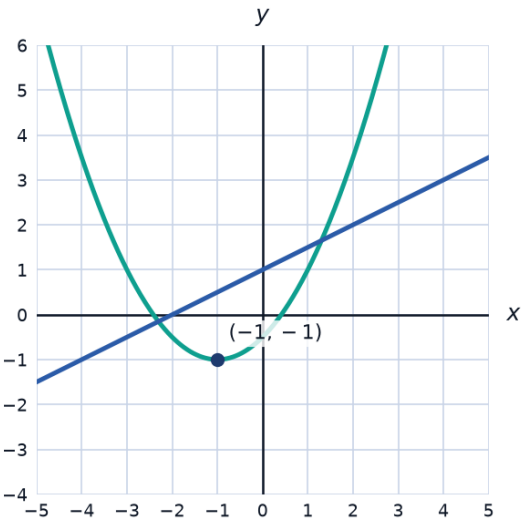
15 **[Challenging]** • **Target: Momentum and Advanced Optimizers** • **Cross-Lesson: 4.01 Conditional Probability and Bayes Rule**

Consider the scalar loss function $L(w) = 5w^2 - 8w + 5$. Find the analytical gradient $abla L(w)$ evaluated at $w = 8$.

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16 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Consider the scalar loss function $L(w) = 3w^2 - 8w + 5$. Find the analytical gradient $ablaL(w)$ evaluated at $w = 4$.



17 [Challenging] • Target: Loss Function Derivatives and Stationary Points • Cross-Lesson: 6.01 Linear Regression and OLS

Consider the scalar loss function $L(w) = 4w^2 - 8w + 5$. Find the analytical gradient $ablaL(w)$ evaluated at $w = 6$.

18 [Challenging] • Target: Multivariate Gradients and Directional Derivatives • Cross-Lesson: 7.01 Logistic Regression and Classification

Consider the scalar loss function $L(w) = 2w^2 - 8w + 5$. Find the analytical gradient $ablaL(w)$ evaluated at $w = 2$.

19 [Challenging] • Target: Gradient Descent Step Updates • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Consider the quadratic loss function $L(w) = w^2 - 4w + 3$. Find the value of w where the function reaches its minimum.

20 [Challenging] • Target: Momentum and Advanced Optimizers • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Consider the following quadratic loss function: $L(w) = 3w^2 - 12w + 10$. What is the value of w that minimizes this loss function? Show your complete algebraic work.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In gradient descent, a weight currently has value $w = 10$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 4$ and the learning rate is $\eta = 0.1$, find the updated weight w'.

a. 10.40

b. 9.60

c. 6.00

d. 9.90

22 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In gradient descent, a weight currently has value $w = 20$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 8$ and the learning rate is $\eta = 0.2$, find the updated weight w'.

a. 12.00

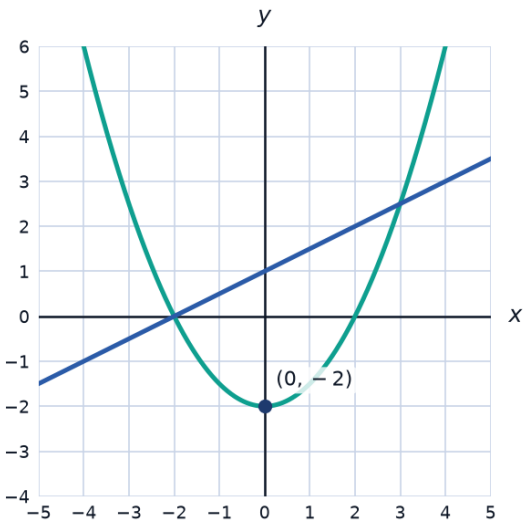
b. 21.60

c. 18.40

d. 19.80

23 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In gradient descent, a weight currently has value $w = 30$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 12$ and the learning rate is $\eta = 0.3$, find the updated weight w'.



a. 18.00

b. 33.60

c. 26.40

d. 29.70

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24 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In gradient descent, a weight currently has value $w = 40$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 16$ and the learning rate is $\eta = 0.4$, find the updated weight w'.

a. 24.00

b. 46.40

c. 33.60

d. 39.60

25 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In gradient descent, a weight currently has value $w = 50$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 20$ and the learning rate is $\eta = 0.5$, find the updated weight w'.

a. 30.00

b. 60.00

c. 40.00

d. 49.50

26 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In gradient descent, a weight currently has value $w = 60$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 24$ and the learning rate is $\eta = 0.6$, find the updated weight w'.

a. 36.00

b. 74.40

c. 45.60

d. 59.40

27 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Consider the scalar loss function $L(w) = 2w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 2$.

a. $\nabla L(2) = 16$

b. $\nabla L(2) = 8$

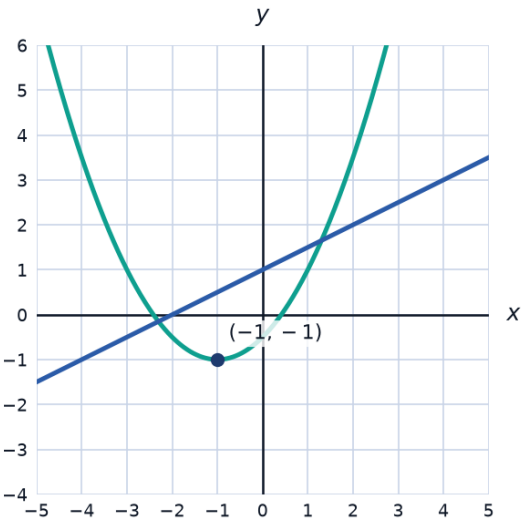
c. $\nabla L(2) = 0$

d. $\nabla L(2) = -4$

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28 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Consider the scalar loss function \(L(w) = 3w^2 - 8w + 5 \). Find the analytical gradient \(\nabla L(w) \) evaluated at \(w = 4 \).



a. \(\nabla L(4) = 16 \)

b. \(\nabla L(4) = 24 \)

c. \(\nabla L(4) = 32 \)

d. \(\nabla L(4) = 4 \)

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Loss Function Derivatives and Stationary Points

Find the stationary point (global minimum) of the scalar mean squared error loss $L(w) = 3(w - 4)^2 + 7$.

a. $w = 4$

b. $w = 7$

c. $w = 3$

d. $w = 0$

30 [Intermediate] • Target: Loss Function Derivatives and Stationary Points

For the loss function $L(w) = w^4 - 4w^2 + 6$, find all values of w where the gradient is zero.

a. $w = 0, \pm\sqrt{2}$

b. $w = \pm 2$

c. $w = 0, \pm 2$

d. $w = \pm 1$

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31 [Intermediate] • Target: Loss Function Derivatives and Stationary Points

In optimization, a critical point has $L'(w) = 0$ and $L''(w) = -8$. What type of point is this?

a. Local Maximum

b. Local Minimum

c. Saddle Point

d. Inflection Point

32 [Intermediate] • Target: Multivariate Gradients and Directional Derivatives

Compute the gradient vector ∇L of $L(w_1, w_2) = 3w_1^2 + 5w_2^2 - 2w_1w_2$ evaluated at $(w_1, w_2) = (2, 1)$.

a. $[10, 6]^T$

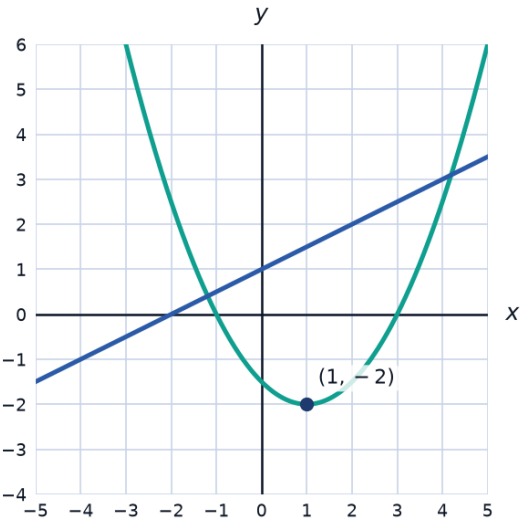
b. $[12, 10]^T$

c. $[8, 8]^T$

d. $[14, 4]^T$

33 [Intermediate] • Target: Multivariate Gradients and Directional Derivatives

In which direction does the negative gradient vector $-\nabla L(\mathbf{w})$ point?



a. The direction of steepest decrease of the loss function

b. The direction of steepest increase of the loss function

c. Orthogonal to the level curves in the direction of highest loss

d. Along the tangent line of the contour

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34 [Intermediate] • Target: Multivariate Gradients and Directional Derivatives

Compute the directional derivative of $f(x, y) = x^2 y$ at $(1, 2)$ in the unit direction $\mathbf{u} = [0.8, 0.6]^T$.

a. 3.8

b. 4.2

c. 2.6

d. 5.0

35 [Intermediate] • Target: Gradient Descent Step Updates

A model weight has value $w_0 = 5.0$. If $\frac{\partial L}{\partial w} = 4.0$ and learning rate is $\eta = 0.1$, find the updated weight w_1.

a. 4.6

b. 5.4

c. 4.0

d. 4.9

36 [Intermediate] • Target: Gradient Descent Step Updates

Consider the quadratic loss $L(w) = \frac{1}{2} a w^2$ with $a = 10$. What is the maximum learning rate η before gradient descent diverges?

a. $\eta < 0.2$

b. $\eta < 0.1$

c. $\eta < 0.05$

d. $\eta < 1.0$

37 [Intermediate] • Target: Gradient Descent Step Updates

Perform one vector gradient descent update from $\mathbf{w}_0 = [1, 2]^T$ with $\nabla L = [2, -4]^T$ and $\eta = 0.25$.

a. $[0.5, 3.0]^T$

b. $[1.5, 1.0]^T$

c. $[0.5, 1.0]^T$

d. $[1.0, 3.0]^T$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Momentum and Advanced Optimizers • Cross-Lesson: 7.01 Logistic Regression and Classification

In Polyak momentum, the velocity update is $\mathbf{v}_{t+1} = \beta \mathbf{v}_t + \eta \nabla L$. If $\beta = 0.9$, $\mathbf{v}_0 = 2.0$, $\eta = 0.1$, and $\nabla L = 4.0$, find $\mathbf{v}_1$.

a. 2.2

b. 2.0

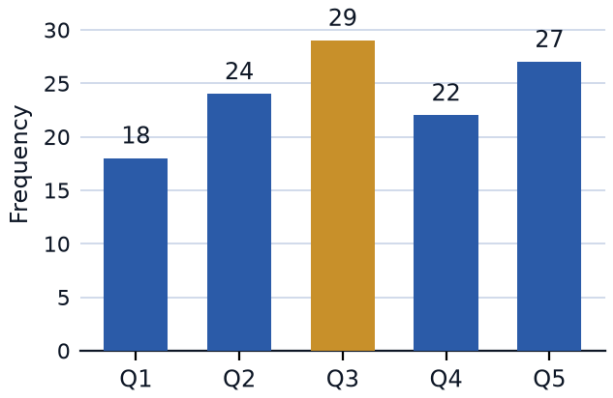
c. 1.8

d. 2.6

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39 [Challenging] • Target: Momentum and Advanced Optimizers • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Why does the Adam optimizer maintain running exponential averages of both first moments (mean) and second raw moments (uncentered variance)?



a. To adapt per-parameter learning rates inversely proportional to gradient magnitude

b. To guarantee convergence to the global minimum in non-convex losses

c. To replace backpropagation with finite differences

d. To eliminate the need for mini-batch sampling

40 [Challenging] • Target: Momentum and Advanced Optimizers • Cross-Lesson: 1.01 Vectors and Embeddings in AI

What is the condition on the Hessian matrix $\mathbf{H} = \nabla^2 L(\mathbf{w})$ for a loss function to be strictly convex everywhere?

a. $\mathbf{H}$ is positive definite everywhere (all eigenvalues > 0)

b. $\mathbf{H}$ has determinant equal to zero

c. $\mathbf{H}$ has trace equal to 1

d. $\mathbf{H}$ is skew-symmetric

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41 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

A 2D loss surface is defined as $L(w_1, w_2) = 6w_1^2 + 11w_2^2$. Find the gradient vector $ablaL$ at the point $(w_1, w_2) = (2, -1)$.

a. $ablaL = [26, -24]^T$

b. $ablaL = [24, 22]^T$

c. $ablaL = [12, -11]^T$

d. $ablaL = [24, -22]^T$

42 [Challenging] • Target: Loss Function Derivatives and Stationary Points • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

A 2D loss surface is defined as $L(w_1, w_2) = 7w_1^2 + 13w_2^2$. Find the gradient vector $ablaL$ at the point $(w_1, w_2) = (2, -1)$.

a. $ablaL = [30, -28]^T$

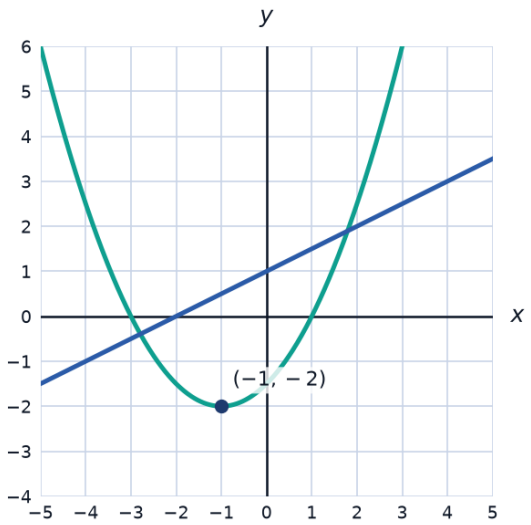
b. $ablaL = [28, 26]^T$

c. $ablaL = [14, -13]^T$

d. $ablaL = [28, -26]^T$

43 [Challenging] • Target: Multivariate Gradients and Directional Derivatives • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

A 2D loss surface is defined as $L(w_1, w_2) = 5w_1^2 + 9w_2^2$. Find the gradient vector $ablaL$ at the point $(w_1, w_2) = (2, -1)$.



a. $ablaL = [22, -20]^T$

b. $ablaL = [20, 18]^T$

c. $ablaL = [10, -9]^T$

d. $ablaL = [20, -18]^T$

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44 [Challenging] • Target: Gradient Descent Step Updates • Cross-Lesson: 6.01 Linear Regression and OLS

A 2D loss surface is defined as $L(w_1, w_2) = 3w_1^2 + 5w_2^2$. Find the gradient vector $ablaL$ at the point $(w_1, w_2) = (2, -1)$.

a. $ablaL = [14, -12]^T$

b. $ablaL = [12, 10]^T$

c. $ablaL = [6, -5]^T$

d. $ablaL = [12, -10]^T$

45 [Challenging] • Target: Momentum and Advanced Optimizers • Cross-Lesson: 7.01 Logistic Regression and Classification

A 2D loss surface is defined as $L(w_1, w_2) = 4w_1^2 + 7w_2^2$. Find the gradient vector $ablaL$ at the point $(w_1, w_2) = (2, -1)$.

a. $ablaL = [18, -16]^T$

b. $ablaL = [16, 14]^T$

c. $ablaL = [8, -7]^T$

d. $ablaL = [16, -14]^T$

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